

Box 1.13. – Memo – Hausdorff convergence

Definition - In a compact metric space X we define the Hausdorff distance on pair of compact subsets of X by setting

$$d_H(A, B) := \max \{ \max \{ d(x, A) : x \in B \}, \max \{ d(x, B) : x \in A \} \}.$$

Properties - We have the following equivalent definition:

- 1) $d_H(A, B) = \max \{ |d(x, A) - d(x, B)| : x \in X \}$
- 2) $d_H(A, B) = \inf \{ \varepsilon > 0 : A \subset B_\varepsilon, B \subset A_\varepsilon \}$, where A_ε and B_ε stand for the ε -neighborhood of A and B , respectively.

Theorem (Blaschke) - d_H is a distance: it is positive and symmetric, it only vanishes if the two sets coincide, and satisfies triangle inequality. With this distance, the set of compact subsets of X becomes a compact metric space itself.

We refer to [13] for a detailed discussion (with proofs) of this topic. Here we only prove a simple fact that we need.

Proposition - If $d_H(A_n, A) \rightarrow 0$ and μ_n is a sequence of positive measures such that $\text{spt}(\mu_n) \subset A_n$ with $\mu_n \rightarrow \mu$, then $\text{spt}(\mu) \subset A$.

Sketch of proof - For each n we have $\int d(x, A_n) d\mu_n = 0$, since μ_n is supported on A_n . Since $d(x, A_n) \rightarrow d(x, A)$ uniformly and $\mu_n \rightarrow \mu$, thanks to the duality between uniform convergence and weak convergence of measures we get $\int d(x, A_n) d\mu_n \rightarrow \int d(x, A) d\mu$. This implies $\int d(x, A) d\mu = 0$ and hence μ is concentrated on A .

Let us also conclude with a result on the stability of the Kantorovich potentials.

Theorem 1.52. *Suppose that X and Y are compact metric spaces and that $c : X \times Y \rightarrow \mathbb{R}$ is continuous. Suppose that $\mu_n \in \mathcal{P}(X)$ and $\nu_n \in \mathcal{P}(Y)$ are two sequences of probability measures, with $\mu_n \rightarrow \mu$ and $\nu_n \rightarrow \nu$. Let (φ_n, ψ_n) be, for each n , a pair of c -concave Kantorovich potentials for the cost c in the transport from μ_n to ν_n . Then, up to subsequences, we have $\varphi_n \rightarrow \varphi$, $\psi_n \rightarrow \psi$, where the convergence is uniform and (φ, ψ) is a pair of Kantorovich potentials for μ and ν .*

Proof. We know that c -concave functions have the same modulus of continuity as c and hence, up to translating by a constant, we can apply the Ascoli Arzelà theorem. We have $\varphi_n \rightarrow \tilde{\varphi}$ and $\psi_n \rightarrow \tilde{\psi}$, the convergences being uniform. From $\varphi_n(x) + \psi_n(y) \leq c(x, y)$ we deduce, as $n \rightarrow \infty$, the inequality $\tilde{\varphi}(x) + \tilde{\psi}(y) \leq c(x, y)$. Moreover we have

$$\mathcal{T}_c(\mu_n, \nu_n) = \int \varphi_n d\mu_n + \int \psi_n d\nu_n \rightarrow \int \tilde{\varphi} d\mu + \int \tilde{\psi} d\nu.$$

Yet, we also have (Theorem 1.50) $\mathcal{T}_c(\mu_n, \nu_n) \rightarrow \mathcal{T}_c(\mu, \nu)$. We infer that the pair $(\tilde{\varphi}, \tilde{\psi})$ is admissible in the dual problem and realizes the maximal value $\int \tilde{\varphi} d\mu + \int \tilde{\psi} d\nu = \mathcal{T}_c(\mu, \nu)$. Hence they are Kantorovich potentials.

The convergence is obviously true on the full sequence (without passing to a subsequence) in case of uniqueness of the Kantorovich potentials at the limit. $\square \quad \square$

1.7 Discussion

This section presents alternative point of views (probabilistic language, polar factorization), connections with economics and finance, generalizations and a brief presentation of the regularity issues on the optimal transport map and the optimal potentials.

1.7.1 Probabilistic interpretation

We completely skipped up to now any interpretations involving probabilities, which are actually natural in this setting. By changing for a while our language, let us describe the situation in the following way.

Two probability measures μ and ν on some spaces (often, $\mathbb{R}^d$) are given, and can be interpreted as the laws of two random variables. Yet, we do not prescribe the joint law (which corresponds to γ) of these two random variables and we consider the optimization problem

$$\min\{\mathbb{E}[c(X, Y)] : X \sim \mu, Y \sim \nu\}$$

where $\mathbb{E}$ denotes the expected value (according to a probability $\mathbb{P}$ on a probability space Ω , which is not relevant for the minimization, but could be considered either “big enough” or to be part of the optimization). This expected value obviously depends on the joint law of (X, Y) , which is the main unknown.

The particular case of real (or vector) valued r.v. with $c(X, Y) = |X - Y|^p$ reads

$$\min\{\|X - Y\|_{L^p} : X \sim \mu, Y \sim \nu\}.$$

More interesting is the case $p = 2$, where the problem can be expressed in terms of covariance. Indeed, let us set $x_0 = \mathbb{E}[X] = \int x d\mu$ and $y_0 = \mathbb{E}[Y] = \int y d\nu$, these two values being the mean values of X and Y . We have

$$\begin{aligned} \mathbb{E}[|X - Y|^2] &= \mathbb{E}[|(X - x_0) - (Y - y_0) + (x_0 - y_0)|^2] \\ &= \mathbb{E}[|X - x_0|^2] + \mathbb{E}[|Y - y_0|^2] + |x_0 - y_0|^2 \\ &\quad + 2\mathbb{E}[X - x_0] \cdot (x_0 - y_0) - 2\mathbb{E}[Y - y_0] \cdot (x_0 - y_0) \\ &\quad - 2\mathbb{E}[(X - x_0) \cdot (Y - y_0)]. \end{aligned}$$

In this expression, the three first terms only depend on the laws of X and Y separately (the first being the variance of X , the second the variance of Y , and the third the squared distance between the two mean values), and the next two terms vanish (since the mean value of $X - x_0$ is 0, and so for the mean value of $Y - y_0$). The problem is hence reduced to the maximization of $\mathbb{E}[(X - x_0) \cdot (Y - y_0)]$. This means that we need to find the joint law which guarantees maximal covariance (i.e. somehow maximal dependence) of two r.v. with given laws. In the case of real valued r.v. (see next chapter) the answer will be that the optimal coupling is obtained in the case where X and Y are completely dependent, and one is an increasing function of the other. The multidimensional case obviously replaces this increasing behavior with other monotone behaviors.

1.7.2 Polar Factorization

A classical result in linear algebra states that every matrix $A \in \mathbb{M}^{N \times N}$ can be decomposed as a product $A = S \cdot U$, where S is symmetric and positive-semidefinite, and U is a unitary matrix, i.e. $U \cdot U^t = I$. The decomposition is unique if A is non-singular (otherwise U is not uniquely defined), and in such a case S is positive definite. Also, one can see that the matrix U of this decomposition is also a solution (the unique one if A is non singular) of

$$\max\{A : R : R \cdot R^t = I\},$$

where $A : R$ stands for the scalar product between matrices, defined as $A : R := \text{Tr}(A \cdot R^t)$.

Indeed, one can write

$$A : R = (S \cdot U) : R = \text{Tr}(S \cdot U \cdot R^t) = \sum_{i,j} S^{ii} U^{ij} R^{ij}$$

where the coordinates are chosen so that S is diagonal (and hence $S^{ii} \geq 0$). Let us observe that $U \cdot U^t = I$ imposes that the vectors $u(i)$ with components $u(i)^j := U^{ij}$ are unit vectors (let us also define the unit vectors $r(i)$ in the same way from the matrix R). Thus

$$\begin{aligned} A : R &= \sum_{i,j} S^{ii} U^{ij} R^{ij} = \sum_i S^{ii} u(i) \cdot r(i) \\ &\leq \sum_i S^{ii} = \text{Tr}(S) = \text{Tr}(S \cdot U \cdot U^t) = \text{Tr}(A \cdot U^t) = A : U. \end{aligned}$$

Also note that maximizing $A : R$ is the same as minimizing $\|A - R\|^2$ (where the norm of a matrix is defined as the square root of its scalar product with itself). Indeed, we have $\|A - R\|^2 = \|A\|^2 - 2A : R + \|R\|^2$ and both $\|A\|^2$ and $\|R\|^2 = \text{Tr}(R \cdot R^t) = \text{Tr}(I) = N$ are constants).

Analogously, in his first works about the quadratic optimal transport, Y. Brenier noted that Monge-Kantorovich theory allowed to provide a similar decomposition for vector fields instead of linear maps.

The statement is the following:

Theorem 1.53. *Given a vector map $\xi : \Omega \rightarrow \mathbb{R}^d$ with $\Omega \subset \mathbb{R}^d$, consider the rescaled Lebesgue measure $\mathcal{L}_\Omega$ on Ω and suppose that $\xi_\# \mathcal{L}_\Omega$ is absolutely continuous; then, one can find a convex function $u : \Omega \rightarrow \mathbb{R}$ and a measure-preserving map $s : \Omega \rightarrow \Omega$ (i.e. such that $s_\# \mathcal{L}_\Omega = \mathcal{L}_\Omega$) such that $\xi = (\nabla u) \circ s$. Moreover, both s and ∇u are uniquely defined a.e. and s solves*

$$\max \left\{ \int \xi(x) \cdot r(x) dx : r_\# \mathcal{L}_\Omega = \mathcal{L}_\Omega \right\}.$$

Note that the statement concerning non-singular matrices exactly corresponds to this one when one takes $\Omega = B(0, 1)$, since the assumption on the image measure corresponds to the non-degeneracy of the linear map $x \mapsto Ax$, the matrix S is the gradient of the convex function $x \mapsto \frac{1}{2}(Sx) \cdot x$ and the unitary matrix U is measure preserving.

We give a proof of this statement, see also [85].

Proof. Consider $\mu = \xi_{\#}\mathcal{L}_{\Omega}$ and take the optimal transport for the quadratic cost between μ and $\mathcal{L}_{\Omega}$. This transport is the gradient of a convex function, that we will call u^* . Set $s := (\nabla u^*) \circ \xi$, and note that $s_{\#}\mathcal{L}_{\Omega} = (\nabla u^*)_{\#}(\xi_{\#}\mathcal{L}_{\Omega}) = (\nabla u^*)_{\#}\mu = \mathcal{L}_{\Omega}$, hence s is measure preserving. Then, take the Legendre transform u of u^* and note $(\nabla u) \circ s = (\nabla u) \circ (\nabla u^*) \circ \xi = \xi$ (indeed, the gradients of both u and u^* are defined a.e. and we always have $(\nabla u) \circ (\nabla u^*) = \text{id}$). In this way we have the desired decomposition.

The uniqueness of ∇u comes from the fact that $\xi = (\nabla u) \circ s$ with s measure-preserving implies $(\nabla u)_{\#}\mathcal{L}_{\Omega} = \mu$ and there is only one gradient of a convex function transporting a given measure (here, $\mathcal{L}_{\Omega}$) to another given measure (here, μ), and it is the optimal transport between them.

Once ∇u is unique, the uniqueness of s is obtained by composing with ∇u^* .

Concerning the optimality of s , use the optimality of ∇u . If we use, instead of the minimization of the quadratic cost the maximization of the scalar product, we get

$$\int (\nabla u(x) \cdot x) dx \geq \int_{\Omega \times \mathbb{R}^d} (y \cdot x) d\gamma(x, y)$$

for every $\gamma \in \Pi(\mathcal{L}_{\Omega}, \mu)$. If we consider a measure-preserving map $r : \Omega \rightarrow \Omega$ and build $\gamma = (r, \xi)_{\#}\mathcal{L}_{\Omega}$, we get

$$\begin{aligned} \int (\xi(x) \cdot r(x)) dx &= \int (y \cdot x) d\gamma(x, y) \\ &\leq \int (\nabla u(x) \cdot x) dx = \int (\nabla u(s(x)) \cdot s(x)) dx = \int (\xi(x) \cdot s(x)) dx, \end{aligned}$$

which provides the optimality of s . □ □

1.7.3 Matching problems and economic interpretations

Optimal transport problems have several “abstract” economic interpretations, where the role transport plans γ is that of matchings between different actors of an economy, and the function $c(x, y)$ does not represent anymore a cost for moving from x to y but rather a compatibility of the two objects x and y or the opposite of the “utility” for x and y to be coupled. Consider that, in economics, it is typical to maximize utility, instead of minimizing costs.

A first easy and well known example is that of the maximization of the productivity. A company has a certain number of employees of different types (let us use the variable x for the types and the measure μ for the distribution of these types, i.e. the quantity of employees for each type) and some tasks to attribute (we use y for the different kinds of tasks, and ν for the distribution of different tasks), and if the productivity $p(x, y)$ of the employees of type x when they work out the task y is known, then the goal of the company is to solve

$$\max \left\{ \int p(x, y) d\gamma : \gamma \in \Pi(\mu, \nu) \right\}.$$

This problem is easy to understand, but we want to analyze some other ones where also the Kantorovich potentials play a role.

The first example is that of stable marriages. Given two populations W of women and M of men, represented by a measure μ on W and a measure ν on M , we denote by $u_w(x, y)$ the interest of Ms x for Mr y , and $u_m(x, y)$ that of Mr y for Ms x . The problem is that of finding a stable set of marriages, i.e. a measure γ on the set $W \times M$ giving how many women of type x get married with how many men of type y . This γ must obviously belong to $\Pi(\mu, \nu)$, and we want to select such a γ so that it is stable, i.e. no new couple (x, y) will decide to divorce (each one from his/her current partner) to go together. The case that we are interested in is that of *transferable utility*: it means that we suppose that, once x and y get married, they decide how to split their total utility $u_w(x, y) + u_m(x, y)$. They divide into a quantity $\phi(x)$, which is the utility surplus for Ms x (now Mrs y , but let us call her with her maiden name) and $\psi(y)$ for Mr y . Note that in this case only the sum $U(x, y) := u_w(x, y) + u_m(x, y)$ really plays a role. A stable marriage is a triple (γ, ϕ, ψ) such that $U(x, y) = \phi(x) + \psi(y)$ γ -a.e. (this represents the fact that ϕ and ψ are a splitting of the total utility) and $U(x, y) \leq \phi(x) + \psi(y)$ for all (x, y) (this represents the fact that no pair (x, y) will be interested in quitting the status quo). It is easy to see that this corresponds to an optimal γ with its potentials. We mention [126] for an interesting discussion of this problem. On the other hand, the case of non-transferable utility is much more involved and does not correspond to a transport problem.

If in the previous example the potentials represented splitted utilities, we can think of other cases where they are really prices. Think indeed that the variable x represents the goods that are available on the market and that μ is the distribution (how many of each type) of these goods, that we consider as fixed. The variable y plays the role of the type of consumers, and ν is their distribution. Let $u(x, y)$ be the utility of the consumer y when he buys the good x . The goal is to determine the prices of the goods and who buys what. Suppose for a while that the price $\phi(x)$ of each good is known; then, each consumer will choose what to buy by solving $\max_x u(x, y) - \phi(x)$. Let us denote (by abuse of notation since usually we used minimization instead of maximization) this quantity as $\phi''(y)$. We describe the choices of the consumers through a measure $\gamma \in \Pi(\mu, \nu)$ where $\gamma(A \times B)$ stands for the number of consumers of type $y \in B$ buying a good $x \in A$. The constraint $\gamma \in \Pi(\mu, \nu)$ stands for the constraints given by the supply and the demand on the market (we say that the market is cleared). Another natural condition to impose is the fact that each consumer only buys goods which are optimal for him, i.e. that γ is concentrated over the set of pairs (x, y) with $\phi''(y) = u(x, y) - \phi(x)$, i.e. such that x is an optimal choice, given ϕ , for y .

This means that we are lead to the following problem

$$\text{find } (\gamma, \phi) \text{ such that } \gamma \in \Pi(\mu, \nu) \text{ and } \phi(x) + \phi''(y) = u(x, y) \text{ } \gamma\text{-a.e.,}$$

and we can see again an optimal transport problem for the maximization of the cost u . Indeed, the pairs (γ, ϕ) can be characterized as the solutions of the Kantorovich problem

$$\max \left\{ \int u \, d\gamma : \gamma \in \Pi(\mu, \nu) \right\}$$

and of the dual problem

$$\min \left\{ \int \phi \, d\mu + \int \phi'' \, d\nu \right\}.$$

By the way, this optimal transport interpretation shows that a simple equilibrium condition (clearing the market and only using rational choices of the consumers) implies good news, i.e. the fact that the general satisfaction $\int u \, d\gamma$ is maximized.

One could remark that the values of the functions φ and φ'' are only defined up to additive constants, which means that this procedure only selects the relative differences of prices between goods, and not the complete price chart. Yet, natural assumptions guarantee uniqueness up to additive constants of the solution of the dual problem (it is the case, for instance, when x and y belong to Euclidean spaces, $u(x, y)$ is differentiable and one of the two measures has strictly positive density a.e.). In this case, as soon as the price of one special good $\bar{x}$ is known, then everything is uniquely determined. A typical example can be obtained if the “empty” good is included in the market. Let us denote by x_0 a special good which corresponds to “not buying anything at all”. We can normalize utilities by setting $u(y, x_0) = 0$ for every y , but this is not really important. However, we can assume that no seller will charge a price different than 0 when selling this empty good. This fixes $\varphi(x_0) = 0$ and allows for computing the other prices. If the measure μ is such that $\mu(\{x_0\}) > 0$, then this means that there are not enough goods on the market so as to satisfy all the demand, and that some of the consumers will stay “out of the market”, i.e. they will buy the empty good.

Another interesting problem is obtained when the measure μ on the sets of goods is not fixed, and we consider a class of goods which are sold by a unique company, acting as a monopolist on this market. We suppose that the set X of feasible goods (those that the company can technically produce) is known, and that it includes the empty good x_0 . The goal of the company is, hence, to select at the same time a measure μ (how much production for each type of goods) and a price list $\varphi : X \rightarrow \mathbb{R}$, satisfying $\varphi(x_0) = 0$, so as to maximize its profit, assuming that each consumer will buy according to the maximization of $x \mapsto u(x - y) - \varphi(x)$.

This problem may be expressed in two ways: indeed, the equilibrium condition of the previous problem (when μ was fixed) induces a relationship between μ and φ , and this allows to consider either one or the other as the variable of the problem.

Probably the easiest way to describe the problem is to think that the company chooses the price list φ , that every consumer y selects its optimal good $X(y) \in \operatorname{argmin}_x u(x, y) - \varphi(x)$, and that the total income (supposing zero production costs) of the company is $\int \varphi(X(y)) \, d\nu(y) = \int \varphi \, d\mu$ for $\mu = X_\# \nu$. The measure μ is also the measure of the real production of goods that the company will implement. In other words, it will adapt its production to the choice of the consumers. Anyway φ has then to be chosen so as to maximize $\int \varphi(X(y)) \, d\nu(y)$, taking into account that the map X depends on φ . The problem can also be adapted so as to take into account production costs, in the form of a function $c : X \rightarrow \mathbb{R}$, and the maximization becomes that of $\int (\varphi - c)(X(y)) \, d\nu(y) = \int (\varphi - c) \, d\mu$. This formulation should be revisited in case the optimal point $X(y)$ is not uniquely defined (since different optimizers for the consumer could lead to very different incomes for the company). We refer to [179] where different relaxed formulations (defined as “optimistic” and “pessimistic”). Anyway, we will see in a while that in a very reasonable case there is no ambiguity.

The other approach is complementary to this one: we can give the company the right to select the production measure μ , then the market is cleared thanks to the previous considerations, thus determining a measure $\gamma \in \Pi(\mu, \nu)$ and a potential φ . The

goal of the company is to maximize $\int (\varphi - c) d\mu$, where μ is the unknown and φ depends on μ . Note that this is an optimization problem in the space of measures which involves the Kantorovich potential φ in the transport from a given measure ν to μ .

The case which is studied the most is the case where both x and y belong to convex sets $X, Y \subset \mathbb{R}^d$ and the function $u(x, y) = x \cdot y$ is a scalar product. This is natural if one thinks that goods can be described through a set of parameters (think at cars: we can use number of seats, size of the luggage van, fuel consumption, maximal speed...) and that consumers are described through the importance that they give to each one of these parameters. Customers of type $y = (y_1, \dots, y_n)$ are ready to spend y_i extra unit of money for every unit increase in the i -th parameter of the good. This means that the values of the coefficients y_i give the relative value of each feature of the good compared to money (in the case of cars, we can expect that they will be higher for richer people, who care less about money, or for people who like a lot, or need a lot, cars). The empty good can be chosen as the point $x_0 = 0 \in \mathbb{R}^d$.

In this case it is easy to check that φ^μ is simply the Legendre transform φ^* of convex analysis. Standard considerations in convex analysis imply that, if φ^* is differentiable at y , then the optimal choice x in $\max_x x \cdot y - \varphi(x)$ is exactly $x = \nabla \varphi^*(y)$ and gives uniqueness a.e. in case ν is absolutely continuous (due to differentiability properties of convex functions). Then, the map $\nabla \varphi^*(y)$ sends ν onto μ and, conversely, $\nabla \varphi$ sends μ onto ν . Hence, the optimization problem reads

$$\max \left\{ \int \varphi_\mu d\mu : \mu \in \mathcal{P}(X) \right\},$$

where, for every μ , φ_μ is the unique convex function with $(\nabla \varphi)_\# \mu = \nu$ and $\varphi(x_0) = 0$. It can also be rephrased as an optimization problem in the class of convex functions, if one takes $\psi = \varphi^*$ as an unknown, since in this case one should maximize $\int \psi^* d(\nabla \psi)_\# \nu = \int (\psi^*(\nabla \psi(y)) - c(\nabla \psi(y))) d\nu$, but we know $\psi^*(\nabla \psi(y)) + \psi(y) = y \cdot \nabla \psi(y)$, which turns the problem into

$$\max \left\{ \int (y \cdot \nabla \psi(y) - \psi(y) - c(\nabla \psi(y))) d\nu(y) : \min \psi = 0, \psi \text{ convex} \right\},$$

the constraint $\min \psi = 0$ coming from $0 = \psi^*(0) = \sup_y 0 \cdot y - \psi(y)$. This is a standard calculus of variations problem, but considered in the restricted class of convex functions. It has been deeply studied in [262], and it is usually known as *principal-agent problem*.

For more general couplings $u(x, y)$ [168] studied fine features on the problem, and in particular which assumptions on the utility u provide a convex optimization problem. Interestingly, the condition to guarantee this convexity is connected to the MTW assumption that we will mention in Section 1.7.6 for regularity.

1.7.4 Multi-marginal transport problems

This recent part of the theory of optimal transport is becoming more and more popular and finds applications in many different fields. We will only give a brief sketch of what is done or can be done on this topic.

It is not difficult to imagine that, instead of using measures $\gamma \in \mathcal{P}(X \times Y)$ with prescribed marginals on the two factors X and Y , one could consider more than two marginals. Take some spaces $X_1, X_2, \dots, X_N$, define $\mathcal{X} := X_1 \times X_2 \times \dots \times X_N$, take a cost function $c : \mathcal{X} \rightarrow \mathbb{R} \cup \{+\infty\}$, some measures $\mu_i \in \mathcal{P}(X_i)$, and solve

$$\min \left\{ \int_{\mathcal{X}} c(x_1, \dots, x_N) d\gamma : \gamma \in \mathcal{P}(\mathcal{X}), (\pi_i)_\# \gamma = \mu_i \right\},$$

where $\pi_i : \mathcal{X} \rightarrow X_i$ is the canonical projection map. Duality could be performed in the same way as for the two-marginal case, getting to problems such as

$$\max \left\{ \sum_i \int_{X_i} \varphi_i d\mu_i : \varphi_i : X_i \rightarrow \mathbb{R}, \sum_i \varphi_i(x_i) \leq c(x_1, \dots, x_N) \right\}.$$

The existence of a transport map (in the sense that there exist maps $T_i : X_1 \rightarrow X_i$, $i \geq 2$, such that the optimal γ is of the form $\gamma = (\text{id}, T_2, \dots, T_N)_\# \mu_1$) is a much more delicate question than the usual case. In some sense, it corresponds to considering the above transport problem as a two-marginal Monge problem on the spaces X_1 and $X_2 \times \dots \times X_N$. Yet, the difference in “dimensions” between the source and target spaces makes the existence of an optimal map trickier. On the other hand, the image measure on $X_2 \times \dots \times X_N$ is not really prescribed, but only its marginals on $X_2, \dots, X_N$ are fixed. This means that extra optimality properties should be taken into account.

This kind of problems may have several different interpretations. For a comprehensive survey about multi-marginal transport problems and their applications in physics and economics, see [249].

Among the first examples that have been treated in this setting we cite the cost

$$c(x_1, \dots, x_N) := \sum_{i,j} |x_i - x_j|^2,$$

which has many similar properties to the 2-marginal case (see [178]), and the determinant cost, where the goal is to maximize

$$c(x_1, \dots, x_d) := \det(x_1, \dots, x_d) \quad \text{or} \quad c(x_1, \dots, x_d) := |\det(x_1, \dots, x_d)|,$$

where the determinant of a family of d vectors in $\mathbb{R}^d$ is intended as the determinant of the matrix where these vectors are the columns (see [110]). The problem is interesting for $d > 2$ since in 2D, from $\det(x, y) = x \cdot Ry$ (where R is the rotation of 90°), one comes back to Brenier problem via a change-of-variable (also compare to **Ex(2)**). The case of [178] will also appear in Section 5.5.5 in connection with barycenters.

Yet, we prefer to present some few cases where important applications naturally lead to a multi-marginal transport problem. For instance, in economics (see [109]), one can think at contracts with more than two agents (when a seller given a good to a buyer in exchange of money, this is a contract between two agents; when somebody wants to build a house, he buys some land and hires a carpenter and an architect, there are at least four agents). A point in the space $\mathcal{X}$ means in this case a possible contract between the agents $x_1, x_2, \dots, x_N$ and $c(x_1, \dots, x_N)$ the global utility of this contract.

A completely different setting is found in recent works from physics, in the framework of Electronic Density Function Theory (see [96] and [129]). To describe in a very simplified way these issues, consider a family of N electrons moving around a nucleus. In quantum physics, the configuration of the electrons is described through a wave function $\psi : \Omega^N \rightarrow \mathbb{C}$, where Ω is the physical region where the electrons live (typically $\Omega = \mathbb{R}^3$). For simplicity, we will ignore spin issues and the fact that the symmetry properties of ψ depend on the distinction between Fermionic and Bosonic particles. We will only consider that the quantity $|\psi|^2$ is invariant under permutations of the variables. This same scalar function denotes the joint law of the position of the electrons. Hence, the probability measure $\rho \in \mathcal{P}(\mathbb{R}^3)$ given by $(\pi_1)_\#(|\psi|^2)$, represents the law of the position of a single electron. The energy of the state is represented by the following sum

$$\hbar \int_{\Omega^N} |\nabla \psi|^2 dx + \int_{\Omega^N} (V(x_1) + \dots + V(x_N)) |\psi|^2 dx + \int_{\Omega^N} \left(\sum_{i < j} \frac{1}{|x_i - x_j|} \right) |\psi|^2 dx,$$

where the first term represents the kinetic energy, the second the potential energy (created, for instance, by the nucleus), and the last the electric interaction energy between the electrons. As far as the parameter N is concerned, it is meant to be the atomic number of an atom, which means that it ranges from 2 to one or two hundreds. The minimization of the above energy becomes a multi marginal transport problem as soon as one takes the semi-classical limit $\hbar \rightarrow 0$ (see [129, 130], the rigorous limit for $N > 2$ is an ongoing work). This allows to get rid of the first term in the energy, and the second is easily seen to depend on ρ only. Hence, the only unknown is the joint law of the positions of the N electrons, i.e. the probability $\gamma \in \mathcal{P}((\mathbb{R}^3)^N)$ given by $\gamma = |\psi|^2 dx_1 \dots dx_N$, with marginals all equal to ρ . The cost c is given by

$$c(x_1, \dots, x_N) := \sum_{i < j} \frac{1}{|x_i - x_j|},$$

so that γ should minimize $\int c d\gamma$ under marginal constraints. The fact that c is neither bounded nor finite (it goes to $+\infty$ as soon as two electrons approach to each other) is a strong difficulty of this problem even in the “easy” case $N = 2$. It is by the way an interesting transport problem where configurations with equalities $x_i = x_j$ cost the most, so that the problem is meaningful and intriguing even and in particular when the marginals are all equal (which would not be the case for costs increasingly depending on the distances).

We observe that we already faced several examples of “symmetric” cost functions, meaningful when the marginals are identical (which is not the case for the costs in [178] and [110]). This raises several questions about symmetric minimizers. Indeed, that all these problems can be restricted to the set of plans γ which are invariant under permutations of the components. The theory of symmetric transport problems, in connections with variant of polar factorization theory and with self-dual convex problems, has been investigated a lot by N. Ghoussoub and collaborators, and we refer to [180, 181] without entering into extra details.

We finish this section by underlining another point: in evolution problems, and particularly in fluid mechanics, a natural Lagrangian point of view is the description

of the global movement by using a measure on the set of possible trajectories. This typically requires the use of probabilities on the set of continuous or Lipschitz paths on $[0, 1]$ but any kind of time-discretization of such a setting can be interpreted in terms of measures on Ω^N , where Ω is the state space and N represents the number of time steps, each one of length $\tau = 1/N$. In this case the marginals are somehow “ordered” and the cost takes into account this feature: for instance, natural costs could be

$$c(x_1, \dots, x_N) = \sum_{i=1}^{N-1} |x_i - x_{i+1}| \quad \text{or} \quad c(x_1, \dots, x_N) = \sum_{i=1}^{N-1} \frac{|x_i - x_{i+1}|^2}{\tau},$$

which are, respectively, the total length of a discrete path or the total kinetic energy (the term $|x - y|^2/\tau$ standing for the integration over a time interval of length τ of a speed $(x - y)/\tau$). Let us mention that the continuous counterpart of this last cost, i.e. $c(\omega) := \int_0^1 |\omega'(t)|^2 dt$ for $\omega \in \mathcal{C} := H^1([0, 1]; \Omega)$ appears in Brenier’s variational formulation of the incompressible Euler equation (see [84]), which gives rise to a minimization problem such as

$$\min \left\{ \int_H c(\omega) dQ(\omega) : Q \in \mathcal{P}(\mathcal{C}), (e_t)_\# Q = \mathcal{L}_\Omega, (e_0, e_1)_\# Q = \gamma_{0,1} \right\},$$

where $e_t : \mathcal{C} \rightarrow \Omega$ is the evaluation map $e_t(\omega) := \omega(t)$, $\mathcal{L}_\Omega$ is the rescaled Lebesgue measure over Ω and $\gamma_{0,1}$ is a fixed measure in $\mathcal{P}(\Omega \times \Omega)$ with both marginals equal to $\mathcal{L}_\Omega$

1.7.5 Martingale optimal transport and financial applications

We present in this section two problems from mathematical finance involving multi-marginal optimal transport. To analyze them, one should know what is an option. An option is a contract giving the owner the right to buy or sell some financial assets at some given conditions, which could bring a monetary advantage. The most typical case is the European Call option, saying “at time T you will have the right to buy this asset at price K ”: should the price be S_T , higher than K , then you gain $S_T - K$; should it be lower, than simply do not use the option, so that finally the gain is $(S_T - K)_+$. More generally, we call option any financial contract giving the owner a gain which is a function of the value S_T at time T of a given asset (a share, for instance). This value is a random variable (based on a probability $\mathbb{P}$) evolving in time. More exotic options such that their payoff depends on the whole history $(S_t)_t$ of the asset value also exist. One of the main issue of financial mathematics is to give formulas to compute the correct price for these contracts. This is based on the no-arbitrage assumption: the only reasonable price for a contract is the one which avoids the existence of arbitrage¹² opportunities on the market, i.e. the possibility of buy-and-sell this contract together with the related underlying asset and to produce a positive amount of money out of nothing (more precisely, since all the values are random variables: to produce with probability 1 a non-negative amount of money, which is strictly positive with strictly positive probability). A general theorem in financial mathematics (see for instance [140] for

¹²The absence of arbitrage is usually refereed to as “no free lunch”.

references) states that all these no-arbitrage prices are actually the expected value of the contract according to a probability $\mathbb{Q}$ which is in general not $\mathbb{P}$ (but $\mathbb{P} \ll \mathbb{Q}$ and $\mathbb{Q} \ll \mathbb{P}$, so that they have the same negligible sets) and which has the property that all the asset values on the market are martingales under $\mathbb{Q}$. This means, for instance, that the price of a European Call option should be $\mathbb{E}^{\mathbb{Q}}[(S_T - K)_+]$. Note that the knowledge of this price for every K implies the knowledge of the law of S_T under $\mathbb{Q}$.

A first problem that one could find in finance is the following: consider a complicated option with payoff depending on the values at time T of several assets S_T^i at the same time T . One should compute $\mathbb{E}^{\mathbb{Q}}[f(S_T^1, S_T^2, \dots, S_T^N)]$. Yet, this depends on the joint law of the different assets under the unknown probability $\mathbb{Q}$. If we suppose that this option is new on the market, but that all the European Calls on each of the assets S^i are regularly sold on the market for each strike price K , then this means that we know the marginal laws μ^i under $\mathbb{Q}$ of each asset, and we only lack the joint law. We can have an estimate of the correct price of the option by computing

$$\min / \max \left\{ \int f(x_1, x_2, \dots, x_N) d\gamma : (\pi_i)_\# \gamma = \mu_i \right\}$$

which is nothing but a transport problem. We ignored here the constraints on the fact that $\mathbb{Q}$ should make the asset value process a martingale since they are already included in the law of each asset and do not involve the joint law. Indeed, in this case the different marginals represent different financial assets, and not different instant of times, and the martingale condition does not play any role¹³.

Another problem, much more intriguing, can be obtained when one considers a single asset, but uses several marginals for several time steps. This allows to consider exotic options depending on the whole history of the asset value. Consider a process $(S_t)_t$ defined for various instants of time (we will consider discrete time models for the sake of simplicity: $t = 0, 1, \dots, N$). Suppose that an option pays $f(S_0, S_1, \dots, S_N)$. Suppose that European Calls are traded in the market for each maturity time t and each strike price K , which provides the law of S_t under the unknown martingale measure $\mathbb{Q}$. Then, the price of the option may be estimated by solving

$$\min / \max \left\{ \mathbb{E}^{\mathbb{Q}}[f(S_0, S_1, \dots, S_N)] : (S_i)_\# \mathbb{Q} = \mu^i, (S_t)_t \text{ is a martingale under } \mathbb{Q} \right\}.$$

This can also be expressed in terms of the measure $\gamma = (S_0, S_1, \dots, S_N)_\# \mathbb{Q}$ as

$$\min / \max \left\{ \int f(x_1, x_2, \dots, x_N) d\gamma : (\pi_i)_\# \gamma = \mu_i, \gamma \in \text{Mart}_N \right\},$$

where the set Mart_T is the set of discrete time martingale measures γ , satisfying for each $t \leq N - 1$

$$\int x_{t+1} \phi(x_0, x_1, \dots, x_t) d\gamma = \int x_t \phi(x_0, x_1, \dots, x_t) d\gamma. \quad (1.7)$$

¹³We could have presented this problem in Section 1.7.4, but we preferred to do it here in order to introduce the martingale setting.

These problems are now called martingale optimal transport problems and are out of the framework of usual transport problems that we presented before. By the way, in general there is no reason to hope for the existence of an optimal transport map. Indeed, even in the easiest case, i.e. two marginals, this will not be the case. The reason is the fact that the only martingale measure which is of the form $\gamma_T = (\text{id}, T)_\# \mu$ is that with $T = \text{id}$. This can be seen from Equation (1.7), since it gives

$$\int T(x)\phi(x) d\mu = \int y\phi(x) d\gamma_T = \int x\phi(x) d\gamma_T = \int x\phi(x) d\mu,$$

which implies, ϕ being an arbitrary test function, $T(x) = x$ μ -a.e. This means that, unless $\nu = \mu$, the martingale transport problem with two marginals admits no admissible transport plan issued by a transport map. On the contrary, it is possible to realize γ as a combination of two transport maps, and optimality results of this kind are proven in [32, 194]. For a general presentation of the martingale optimal transport problem, see [33] and [174].

We finish this section presenting an alternative approach, leading to the same problems, and based on the idea of optimal hedging. Think that $f(S_0, \dots, S_N)$ is the loss that a company would face, depending on the prices of the asset S (i.e. it is the amount of money that the company needs to pay, because of a debt or other contracts). The company wants to “cover” (or “hedge”) this amount, by using the possibilities in the market. In order to hedge this amount, the company can

- buy usual options, based on the value of the asset at time t , i.e. buy a contract whose value will be $\phi(S_t)$, and whose price is known to be $\int \phi d\mu_t$;
- buy some amounts of the asset itself: at time $t \leq N-1$, it is possible to buy for a price S_t and to re-sell at price S_{t+1} ; the number of assets to buy can only be chosen according to the information that are available at time t , which means that one can buy (and then re-sell) an amount of the form $\psi(S_0, S_1, \dots, S_t)$; this results in covering a quantity $\psi(S_0, S_1, \dots, S_t)(S_{t+1} - S_t)$, at cost 0.

The optimal hedging problem becomes

$$\min \left\{ \sum_{t=0}^N \int \phi_t d\mu_t : \sum_{t=0}^N \phi_t(x_t) + \sum_{t=0}^{N-1} \psi(x_0, x_1, \dots, x_t)(x_{t+1} - x_t) \geq f(x_0, x_1, \dots, x_N) \right\}.$$

It is not difficult to check, thanks to the marginal constraints and to the martingale condition (1.7), that this problem is exactly the dual of the above martingale transport problem.

1.7.6 Monge-Ampère equations and regularity

We saw in Section 1.3.1 that the optimal transport for the quadratic case is of the form $T(x) = \nabla u(x)$, for a convex function u . If this function is smooth and strictly convex, and the two measures μ and ν are absolutely continuous (say, with densities f and g) then we can write the condition $T_\# \mu = \nu$ in a PDE form. Indeed, this condition is

equivalent, if $\mu = f(x) dx$ and $\nu = g(y) dy$, and if T is C^1 and injective, to the Jacobian condition

$$\det(DT(x)) = \frac{f(x)}{g(T(x))}$$

which can be obtained as a consequence of a simple change-of-variable computation, as explained here below.

Box 1.14. – Memo – Change-of-variable and image measures

Proposition - Suppose that $\rho \in L^1(\Omega)$ is a positive density on $\Omega \subset \mathbb{R}^d$ and $T: \Omega \rightarrow \mathbb{R}^d$ is a Lipschitz injective map, which is thus differentiable a.e. We suppose that $\det(DT) \neq 0$ a.e. on $\{\rho > 0\}$. Then the image measure $T_{\#}\rho$ is absolutely continuous and its density ρ_T is given by

$$\rho_T(y) = \frac{\rho(T^{-1}(y))}{\det(DT(T^{-1}(y)))}.$$

If T is non-injective, the formula becomes $T_{\#}\rho = \rho_T \cdot \mathcal{L}^d$ with ρ_T given by

$$\rho_T(y) = \sum_{x: T(x)=y} \frac{\rho(x)}{\det(DT(x))}.$$

The same formulas stay true if T is not Lipschitz, but if there are measurable sets E_i such that $|\Omega \setminus \bigcup_i E_i| = 0$ and T is Lipschitz continuous on each set E_i (see also Definition 3.13 for countably Lipschitz functions), with the differential DT which is actually the differential of the restriction of T to each set where it is Lipschitz continuous (and coincides thus with the approximate differential of T , see Section 3.3.2).

In the particular case where $T = \nabla u$ (where we suppose u to be strictly convex just to guarantee injectivity of T), this becomes the so-called Monge-Ampère equation

$$\det(D^2 u(x)) = \frac{f(x)}{g(\nabla u(x))}, \quad (1.8)$$

which is a nonlinear PDE of elliptic type. Nonlinearity comes both from the right-hand side with the term $g(\nabla u)$ but also from the higher order term $\det(D^2 u)$. This Jacobian term is non linear, but has some monotonicity property when we assume the matrix $D^2 u$ to be positive (which is the case for convex functions), as we have $0 \leq A \leq B \Rightarrow \det(A) \leq \det(B)$. This monotonicity is anyway very much degenerate, but enough to classify this non-linear PDE as a fully nonlinear (degenerate) elliptic equation. We refer, for instance, to [191] for the general theory about this equation.

In our case, it is important to understand the meaning and the boundary conditions to be associated to this equation.

First, let us consider the boundary conditions. In view of the transportation meaning of this PDE, if we consider $\mu \in \mathcal{P}(\Omega)$ and $\nu \in \mathcal{P}(\Omega')$, there is no point in imposing the behavior of u or ∇u on $\partial\Omega$. On the contrary, the natural condition is just $\nabla u(\Omega) = \Omega'$, which is a global condition on u . This corresponds, roughly speaking and only when u

is a homeomorphism, to $u(\partial\Omega) = \partial\Omega'$. This particular sort of “boundary” condition is usually called *second boundary value problem* for the Monge-Ampère equation.

We define various notions of solutions for (1.8):

- we say that u satisfies (1.8) in the Brenier sense if $(\nabla u)_\#(f \cdot \mathcal{L}^d) = g \cdot \mathcal{L}^d$ (and this is actually the sense to be given to this equation in optimal transport);
- we say that u satisfies (1.8) in the Alexandroff sense if the measure ρ defined via $\rho(A) := |\bigcup_{x \in A} \partial u(x)|$ is absolutely continuous and its density equals the right-hand side of (1.8) a.e. (warning: the fact that ρ is a measure is not straightforward, one has to use properties of convex functions, as in Lemma 1.1.12 and Theorem 1.1.13 in [191]);
- we say that u satisfies (1.8) in the viscosity sense if it satisfies the usual comparison properties required by viscosity theory but restricting the comparisons to regular convex test functions (this is possible because of the monotonicity properties of the determinant);
- we say that u satisfies (1.8) in the classical sense if it is of class C^2 and the equation holds pointwise.

Note that all these notions except the first may be also applied to the more general equation $\det D^2 u = f$ (for fixed f , without the structure $f/g \circ T$), while the first one just applies to this specific transportation case.

The regularity of the solutions of (1.8), which implies regularity results for the optimal transport map, has been studied in the '90s by Caffaelli, with very strong results. The results we want to use are well summarized in Theorem 50 of [292]:

Theorem 1.54. *If f and g are $C^{0,\alpha}(\Omega)$ and are both bounded from above and from below on the whole Ω by positive constants and Ω is a convex open set, then the unique Brenier solution u of (1.8) belongs to $C^{2,\alpha}(\Omega) \cap C^{1,\alpha}(\overline{\Omega})$ and u satisfies the equation in the classical sense (hence also in the Alexandroff and viscosity senses).*

We just detail a possible bibliographical path to arrive at this result. It is not easy to deal with Brenier solutions, so the idea is to consider viscosity solutions, for which it is in general easy to prove existence by Perron's method. Then prove some regularity result on viscosity solutions, up to getting a classical solution. After that, once we have a classical convex solution to Monge-Ampère equation, this will be a Brenier solution too. Since this is unique (up to additive constants) we have got a regularity statement for Brenier solutions. We can find results on viscosity solutions in [99], [101] and [100]. In [99] some conditions are given so as to ensure strict convexity of the solution of $\det D^2 u = f$ when f is bounded from above and below. In [101] for the same equation it is proved $C^{1,\alpha}$ regularity provided we have strict convexity. In this way the term $f/g(\nabla u)$ becomes a $C^{0,\alpha}$ function and in [100] it is proved $C^{2,\alpha}$ regularity for solutions of $\det D^2 u = f$ with $f \in C^{0,\alpha}$.

Some recent improvements on this regularity theory are due to De Philippis and Figalli, who worked on this natural question: if f and g are bounded from above and below, then the transport map is a BV map (since it is the gradient of a convex function), which happens to be continuous (since $u \in C^{1,\alpha}$); its derivative has no jump part

(because of continuity), and the question arises whether $T \in W^{1,1}$, i.e. $u \in W^{2,1}$. The answer is yes, as it is shown in [148, 150, 151], and one can even prove sharper estimates (D^2u is not only L^1 but slightly better, including some $L^{1+\varepsilon}$ results, also proven by Schmidt in [281]).

The situation becomes much trickier when dealing with different costs than the quadratic one. If the cost c satisfies the twist condition it is still possible to express the optimal map T in terms of a gradient, $T(x) = (\nabla_x c(x, \cdot))^{-1}(\nabla \varphi(x))$. If we write $\nabla_x c(x, T(x)) = \nabla \varphi(x)$, differentiating this equality we get

$$D_{xx}^2 c(x, T(x)) + D_{xy}^2 c(x, T(x)) DT(x) = D^2 \varphi(x),$$

which means that the equation on $\det(DT)$ reads

$$\det(D_{xy}^2 c(x, T(x))) \det(DT(x)) = \det(D^2 \varphi(x) - D_{xx}^2 c(x, T(x))),$$

i.e.

$$\det(D^2 \varphi(x) - D_{xx}^2 c(x, T(x))) = \det(D_{xy}^2 c(x, T(x))) \frac{f(x)}{g(T(x))},$$

which is a Monge-Ampère-type equation of the form

$$\det(D^2 \varphi(x) - A(x, \nabla \varphi(x))) = F(x, \nabla \varphi(x)).$$

The regularity problem of the Kantorovich potentials for general costs has been a long-standing problem, and it was listed as open in [292]. It is now well clarified after a remarkable paper by Ma, Trudinger and Wang, who found out in [220] a key assumption on the cost c so as to be able to develop regularity estimates, based on some Pogorelov-type methods. The assumption on c is very technical and requires some inequalities on its fourth-order derivatives. We do not want to write them here, but it is worth noting that these assumptions have later been proven to be sharp by Loeper in [217]. In particular, the case $c(x, y) = |x - y|^2$ is covered by this assumption but is a limit case, and no other power cost $|x - y|^p$ for $p \in]1, +\infty[$ satisfies it. The cost $\sqrt{\varepsilon^2 + |x - y|^2}$ also satisfies the MTW assumption for every $\varepsilon > 0$, and this allowed some approximation result for the linear cost $|x - y|$ (see [211]). The convexity of the domain Ω is also replaced by a suitable notion of c -convexity that we do not detail here. The connection with convexity properties related to c -concave functions is also underlined in [168], as we saw in Section 1.7.3. The theory has been deeply studied after this discovery and many costs have been considered: the case of the squared distance on a manifold can also be investigated, and the condition translates into a curvature condition on the manifold. We refer to [165] for an interesting survey of the whole theory.

Chapter 2

One-dimensional issues

This chapter is devoted to the special features of the 1D case, and to a less known transport map in the higher-dimensional case, called the *Knothe map*, built from 1D constructions and related to some degenerate transport cost. The discussion section deals with applications of 1D transport maps to image processing and geometric inequalities.

2.1 Monotone transport maps and plans in 1D

We already discussed, in Section 1.3, the quadratic cost in 1D: as soon as the source measure μ has no atoms, there exists an optimal map, which is monotone non-decreasing.

In this section we want to discuss monotone (in the following “monotone” means “monotone non-decreasing”) transport maps between two given measures in terms of their cumulative distribution functions, and generalize to the case where μ may have atoms. This discussion is independent of optimal transport considerations.

Box 2.1. – Memo – Cumulative distribution function

Definition - Given a probability measure $\mu \in \mathcal{P}(\mathbb{R})$ we define its cumulative distribution function F_μ through

$$F_\mu(x) = \mu((-\infty, x]).$$

The cumulative distribution function F_μ is easily seen to be non-decreasing and right-continuous since if $x_n \rightarrow x$ with $x_n > x$, then $(-\infty, x] = \cap_n (-\infty, x_n]$ and hence $\mu((-\infty, x]) = \lim_n \mu((-\infty, x_n]) = \inf_n \mu((-\infty, x_n])$. It is continuous at any point where μ has no atom since if $x_n \rightarrow x$ with $x_n < x$, then $\lim_n \mu((-\infty, x_n]) = \mu((-\infty, x])$.

The CDF F_μ is important because it characterizes the measure μ . Indeed, the sets of the form $(-\infty, x]$ are enough to generate all open sets: we have $\mu([a, b]) = F_\mu(b) - F_\mu(a)$, $\mu(]a, b[) = \sup\{F_\mu(t) - F_\mu(a) : t < b\}$ and the measure of any open set is obtained via countable disjoint unions of open intervals.

In $\mathbb{R}^d$ the situation is trickier, but the knowledge of $\mu([-\infty, x_1] \times \cdots \times [-\infty, x_d])$ for every $x = (x_1, \dots, x_d)$ is enough, as in 1D, to characterize μ (by finite differences one could